

Analytics to Action

Participatory Data Design

Anders Koed Madsen, March, 2026

Theme 2: Making Data Valuable

How to make data and data analysis valuable to different stakeholders in practice.

Three sessions

1. Datafication
2. Exploratory data analysis
- 3. Participatory data design**

Readings for today

- Jensen, T. E., Birkbak, A., Madsen, A. K., & Munk, A. K. (2021). Participatory Data Design: Acting in a digital world. In *Making & Doing: Activating STS through Knowledge Expression and Travel* (pp. 117-136). MIT press.
- Madsen, A. (2024). Digital methods as 'experimental a priori'—how to navigate vague empirical situations as an operationalist pragmatist. *Convergence*, 30(1), 94-115.

Agenda for this lecture

- Definition: Participatory Data Design.
- Involving stakeholders to align data projects and make them relevant
- Gephi Lite as flexible visualization
- Involving stakeholders to challenge their assumptions and generate hypotheses

BREAK

- Niels from W&A Visits (10.30-10.50)
 - Students working with Rigshospitalet can leave and wait to join their call with Maja at 11.00.

Participatory Data Design

Definition

Participatory Data Design (PDD)

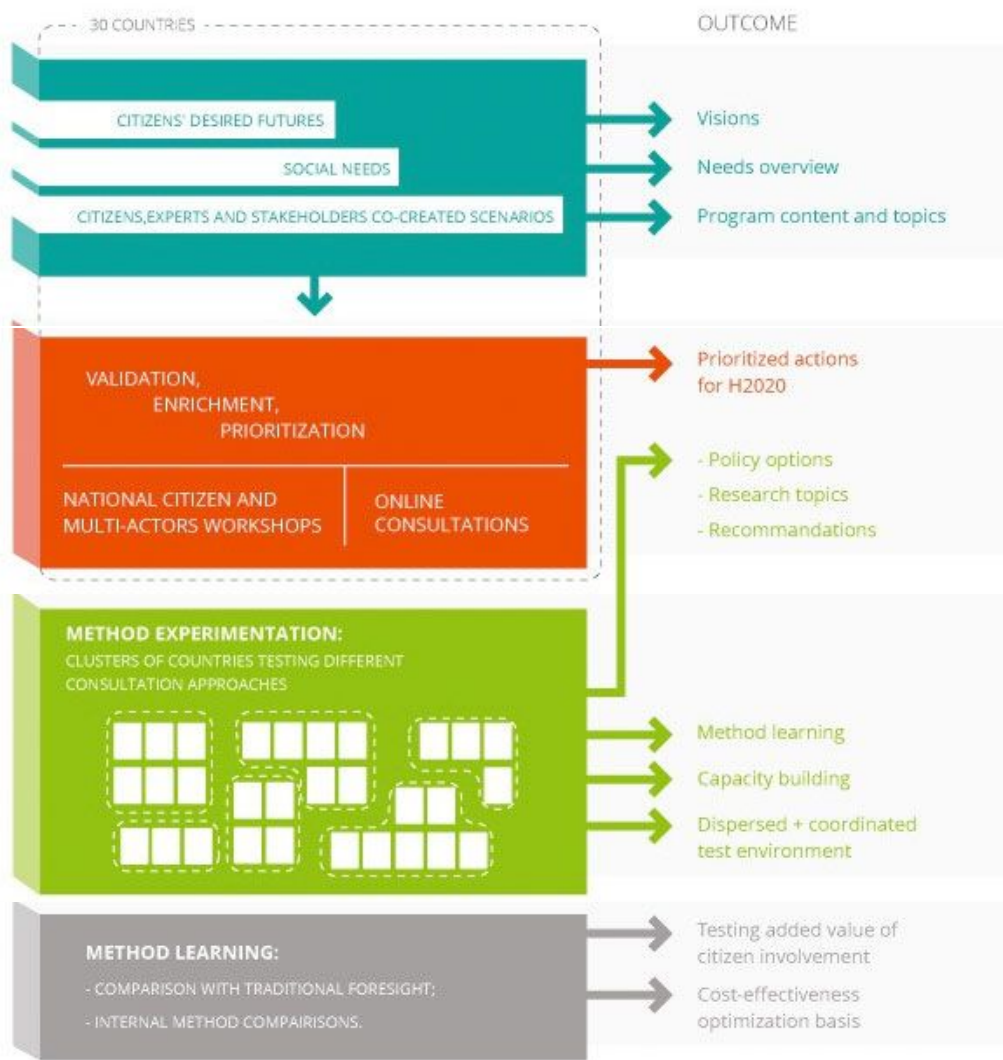
- PDD refers to the **involvement of stakeholders in data science** projects.
- **Involvement**: active participation, typically upstream, influencing choices relating to datafication, analysis, and/or presentation.
- **Upstream** because critical decisions that decide whether input and concerns can be taken into account are otherwise taken.
- **Active participation** because challenges and opportunities will otherwise remain tacit and unacknowledged by the stakeholders.
- Often happens through **data sprints**, i.e. workshops of varying length where pilot projects are co-developed and intermediary, exploratory visualizations are used to elicit running feedback from stakeholders.

Involving stakeholders to make a data project relevant and align with their interests

(“Having data as a boundary object”)



Prioritizing Zika virus stories from Twitter for public deliberation with the ASSET and CIMULACT projects



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CREATING
SOCIETY
TOGETHER



Racy
@RacySicilian



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WOW!! Homeland Security Not Screening New Arrivals for Zika Virus

[breitbart.com/national-secur...](http://breitbart.com/national-security)

#PJNET
#TCOT
#Zika



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27

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16



5:40 PM - 14 Feb 2016



Hot Patriot Chick
@HotPatriotChick



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#Ccot #rednationrising #pjnet #uppusa
#ZikaVirus #Zikaepidemic #vaccines
#vaccinateNOW #Zika



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2

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1



2:41 PM - 20 Feb 2016



CitizenGO
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U.S. Congressmen scolds @UN rights chief for exploiting the #zika virus to push #abortion.
lifesitenews.com/news/us-congre...



TRU NEWS™
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"I Have #Microcephaly. We Need Treatment, Not #Abortion" tru.news/1Q5FggO #zika #prolife



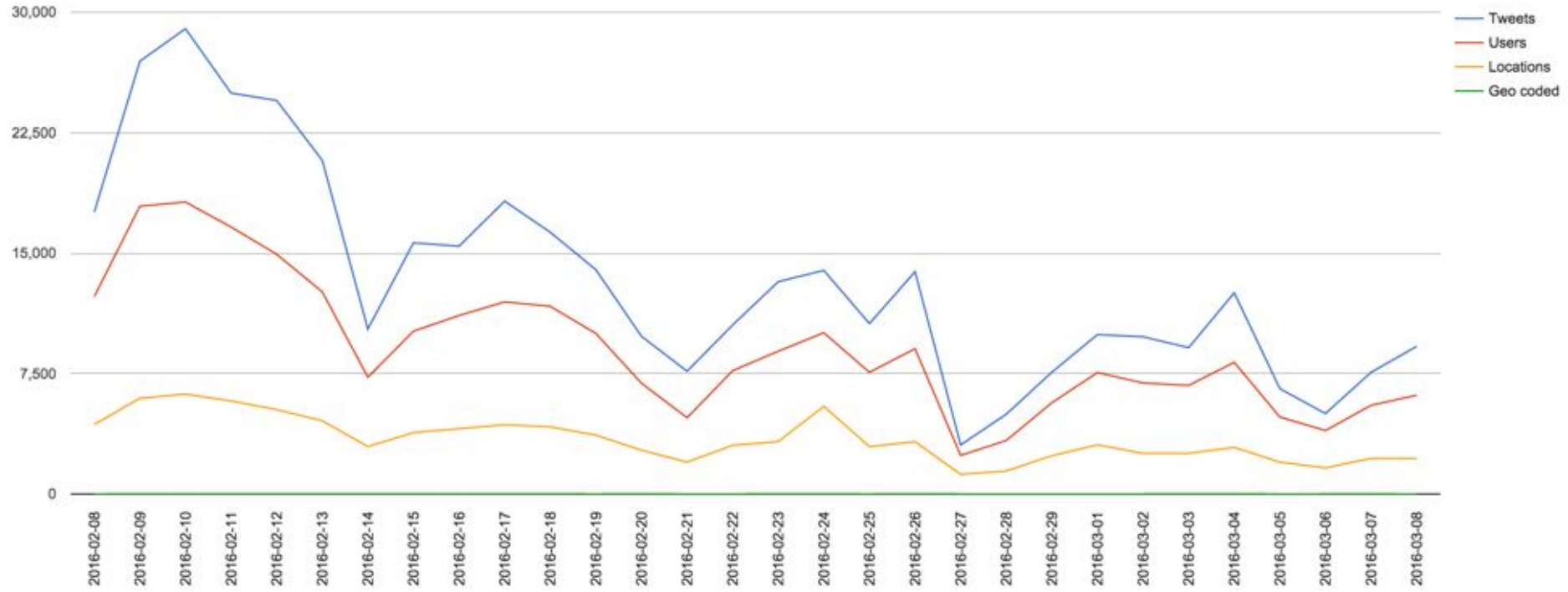
RETWEETS
9

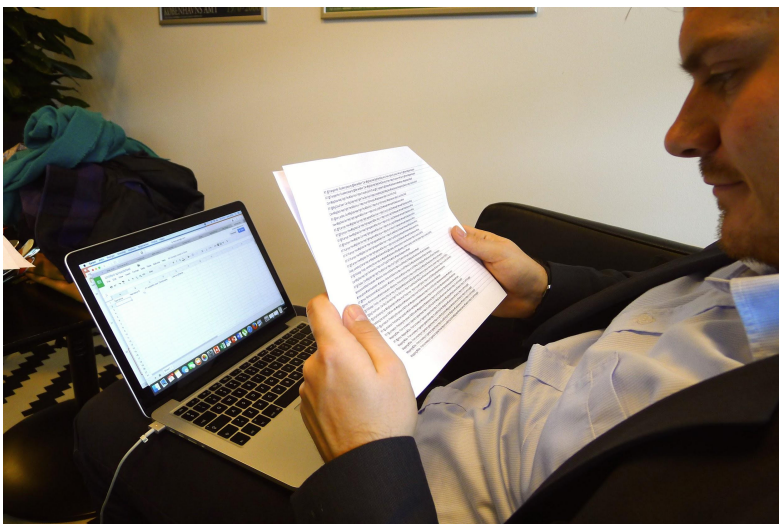
LIKE
1



11:00 PM - 9 Feb 2016

400,000 tweets containing 2,500 hashtags in a month





◀ Identifying narratives through co-hashtag analysis

Safari Arkiv Rediger Oversigt Historie Bogmærker Vindue Hjælp

docs.google.com

Lukas Graham - Red Wi... Mail - John Haukeland ... John Score - Google Sh... Twitter / Notifications Zika_narratives - Google... Untitled presentation ... Tag screenshot med M... jhaukeland@gmail.com

John Score

File Edit View Insert Format Data Tools Add-ons Help All changes saved in Drive

fx 4

	A	B	C	D	E	F	G	H	I	J
	Narrative	Narrative #	John Score	Controvers	Link to q/then	Multiplicity kild		Theme r		
1	Zika Christian religion	1	1	2	2			V	Conservative Ch	Top Conservatives On
2	Contraception Pope Gender	3	4	4	4	4		V		Patri
3	Growing Support Among Experts for Zik	4	3	3	4	4		V	Gender issue, planned familyhood, high	
4	Rio OL	6	2.5	4	3	4		Trust/Governanc	Free movement, Rio	
5	Illegals (Obama)	9	1	2	2			Trust/Governanc	Link to freedom of movement, but basic	
6	Homeland Security Not Screening New	10	2	3	2	2		Governance	Risk communication, retrict/control mov	
7	IAEA to assist Latin American	11	2	3	2.5	2		Trust/Governanc	Link til privacy og brug af early detector	
8	Google big data zika	13	3.5	3	4			Governance	Zika er en ondt sygdom. En vaccine ville	
9	Suspects muscito insecticide to cause bi	15	1	2	1			Trust	Risk communication and parallel inform	
10	Negative kosekvenser ved vaccine	17	1	1	2			Trust	Vrede mænd, der mistænker bivirkninge	
11	Sexual trans of Zika	18	1	3	2			Health Lit	Ingen diskusison bare et empirisk stater	
12	30secmoms	19	1.5	1	3			Health Lit	Breastfeeding in the context of Zika viru	
13	What is #Zika	20	1.5	1	3			Health Lit	Information om symptomer fra Pan Ameri	
14	BioDiv is key to control Zika	22	2.5	2	3	3		Health Lit/Goven	Link til Governance og andre emner =>	
15										
16										
17										
18										

Sheet1

◀▲ Developing the “John Score”

Jensen, T. E., Birkbak, A., Madsen, A. K., & Munk, A. K. (2021). Participatory Data Design:

Alexis López

@Dalexlo



Follow

Can #BigData Help Fight The #Zika Virus?
ow.ly/2bwQt8 #DataScience



What is important here?

- Allowing stakeholders to participate in the definition of what is interesting.
- Keeping decisions about datafication and analysis as open as possible.
- Using intermediary visualizations as concrete occasions for stakeholders to provide feedback.

5 minutes in groups: What could be a minimum viable way to involve your case company at your presentation?

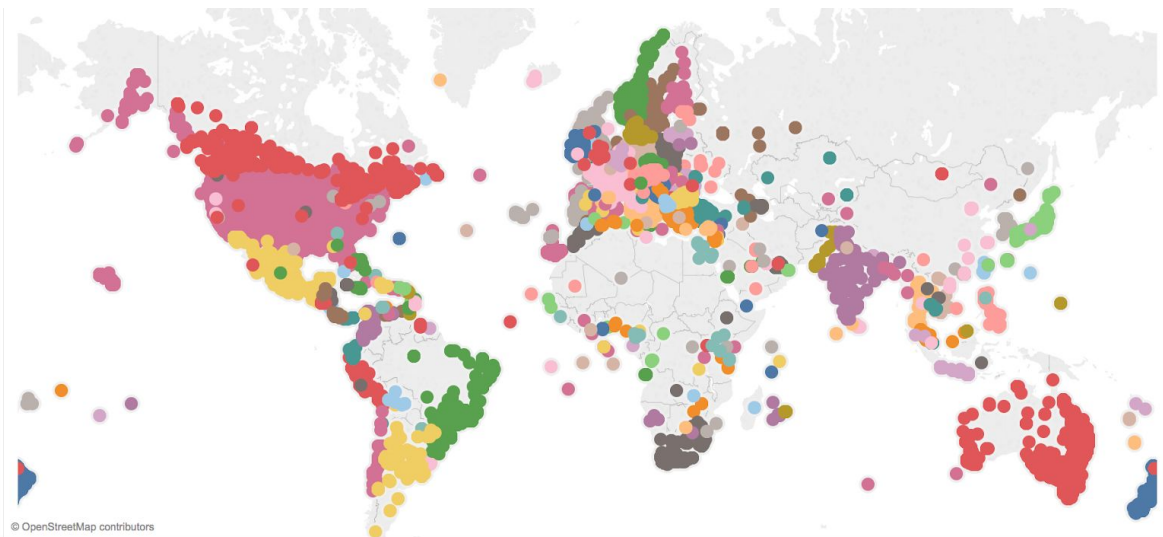
Flexible (network) visualizations

“visualization can do some of the same work that cheap materials such as cardboard boxes and Lego brick models did in the Scandinavian tradition of participatory design”

“[...] occasionally the participants would start some kind of operation that would cause the networks on the screen to slowly start moving as if the whole network was being pulled apart while some parts of it were still hanging together. Seconds later, the participants might stop the movement, study the new configuration carefully, and perhaps add color, zoom in, or in some other way manipulate the display (Jensen et al, 2021)”

DEMO

Working with flexible visualization to
understand the global market of
professional chefs

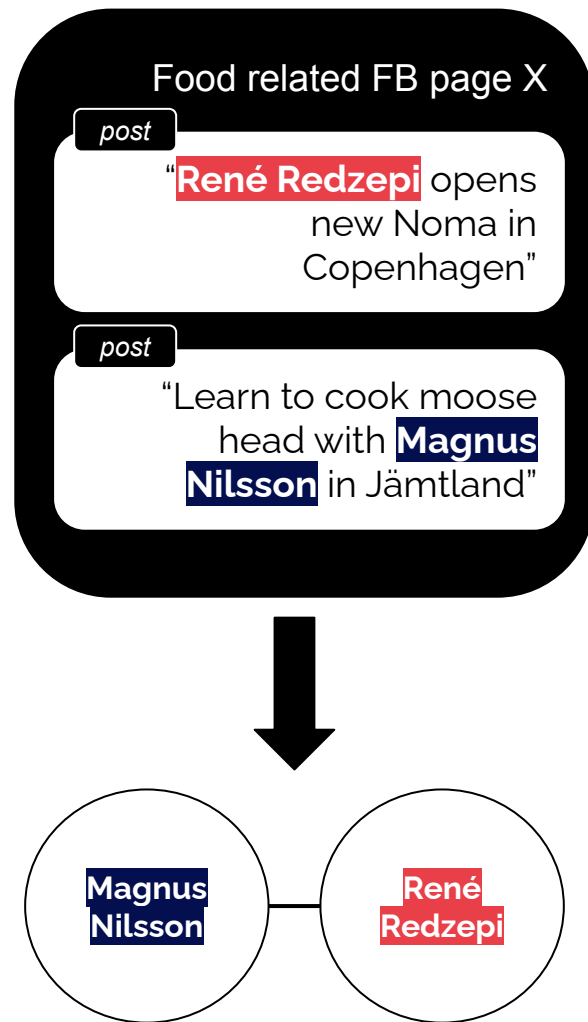


Dataset: All posts (text) from 242.000 food-related Facebook pages worldwide between 2010 and 2018

+ 670 names of famous chefs (scraped from Wikipedia)

NETWORK

Nodes = chef names
Edges = co-mention on food-related page



Variables on nodes

- **Country names:** To what extent has a chef been talked about by pages from this country.
- **Nordic:** Is the chef from one of the nordic countries? If yes, which?
- **Mention count:** How many times is the chef mentioned?
- **Page count:** How many different pages are mentioning this chef?
- **Country count:** How many different countries are the pages mentioning this chef from?
- **Degree:** How many chefs other chefs is this chef mentioned together with?
- You can also use the statistics module to compute the modularity class (using the Louvain method), i.e. cluster the chefs based on the topology of the network.

Variables on edges

- **Raw overlap:** How many pages are co-mentioning two chefs?
- **Normalized overlap:** How many pages are co-mentioning two chefs as a proportion of the max possible given as the number of pages mentioning each chef?
- **Only nordic overlaps:** If only pages from the nordic countries are taken into account as co-mentioning chefs.

**Let's look at the network as an example
of a flexible visualization to be used in
PDD processes.**

Work after class

Download the .gexf file from Learn (called “chef_overlap_full)

Analyse it in Gephi Lite: <https://gephi.org/gephi-lite/>

Try to engage with the data:

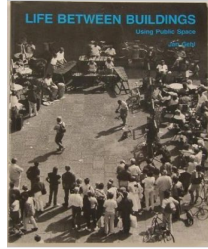
- to find patterns that seem interesting in order to understand the global market of chefs
- to list who you would ideally invite in to help you explore this data. What type of ‘domain expertise’ would you need to ensure a good interpretative process?

**Involving stakeholders to challenge their
preference for agreement and explore
new hypotheses**

(Case: GEHL)

Gehl

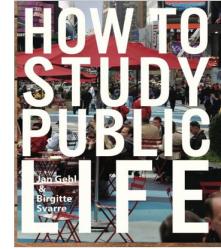
"We measure what we care about"



1971



2010



2013



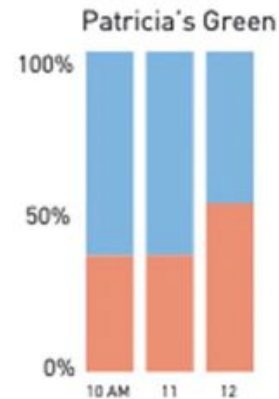
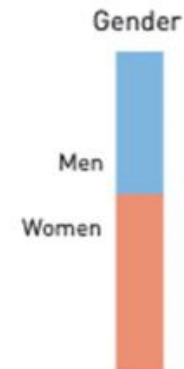
2020



The problem: urban political diversity



'The Gehl Lens' (The public life diversity toolkit)



“The Facebook lens”



An example of a Facebook profile that self-identifies as 'political' is the official site for the Danish Prime Minister, Mette Frederiksen



As Mette Frederiksen is member of the Social Democrats ('A') all her posts are taken to express statements supporting the red block in Danish politics.



One such post is this statement supporting the need for funding research in sustainable fuels



Users who place more than 60% of their positive emojis within one block is considered supportive of that block



The outcome of this process is a list of 600.000 user ID that each have a color based on their political leanings

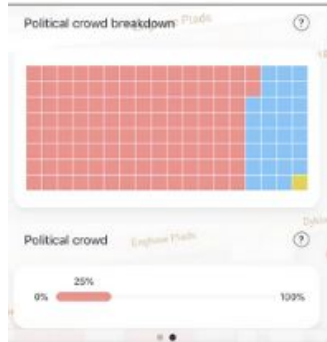
“The Facebook lens”



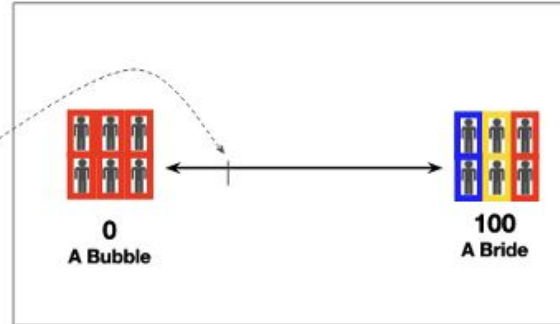
An example of a Facebook event in Copenhagen is this concert by Bob Hund which have more than 1000 users showing interest

8429386frnufhpe	8429386frnufhpe
8h74gf8y4fh807hf	8h74gf8y4fh807hf
3480932jfklnrfvf	84hehfjrhpoe8ho
209438463ejbkfbkd	84hfæehhå48fep
85urhd74rjnr	85urhd74rjnr
84i4jue7ei4oeke	84i4jue7ei4oeke
93843hreolng	94ieju8er8eiek
09580y95ujglkr	94irjnr8eiekdorf
455jnduririrjko94	455jnduririrjko94
.....	

The list of user IDs attending vs. The list of political IDs



Using the list comparison to visualize the political crowd and it's block-break down



Translating the breakdown into a diversity index for each event.

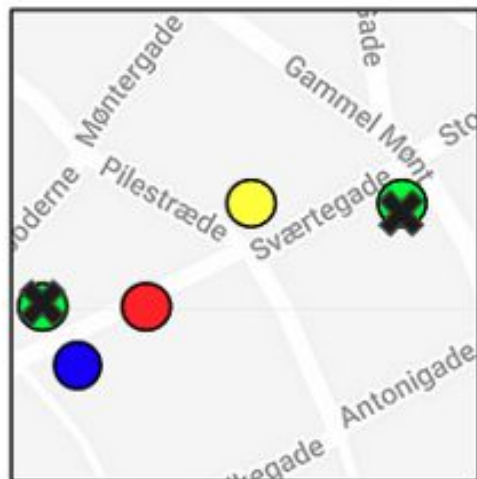
Do you live in a bubble?

A datascape on urban political diversity by

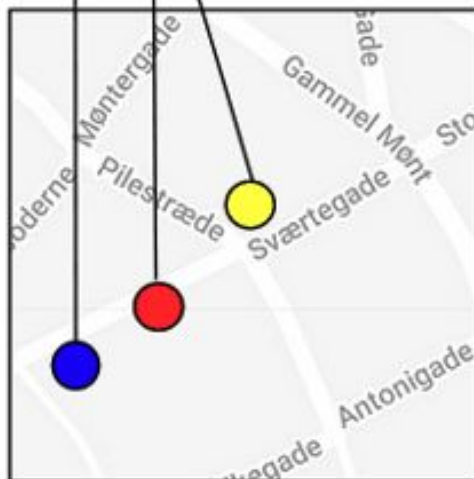


and

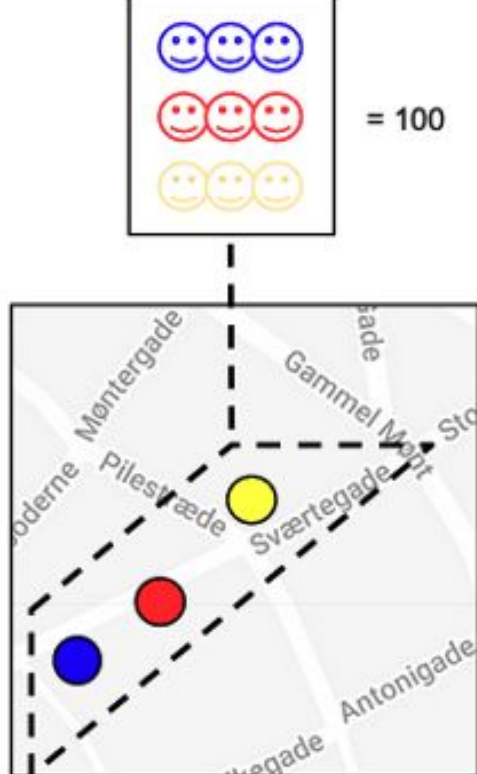
Gehl



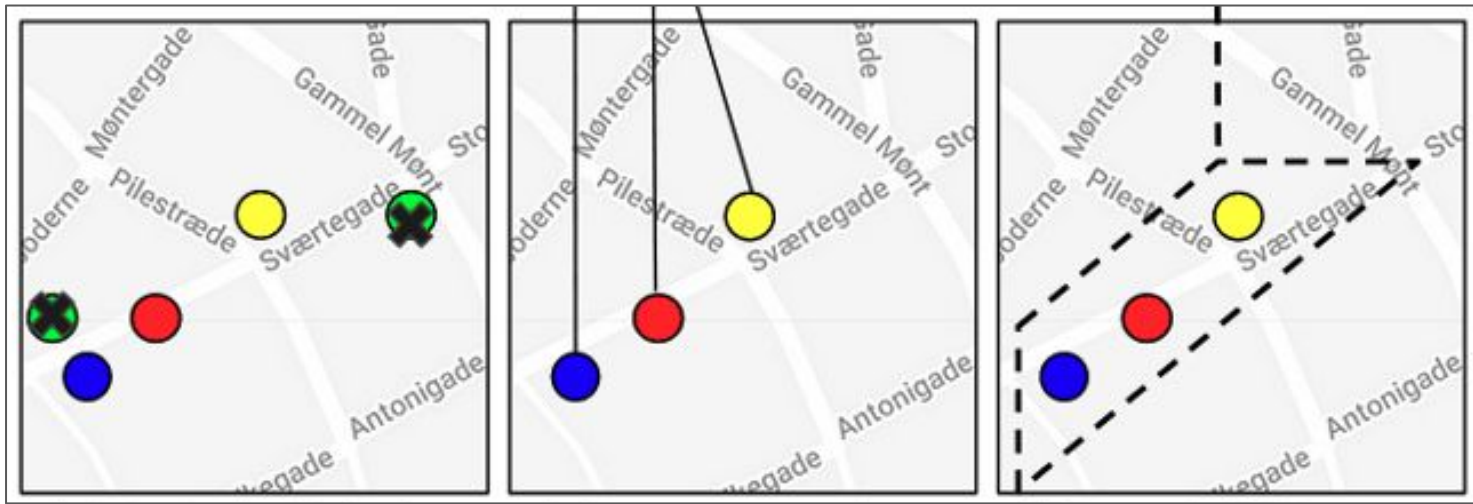
"I think we should only look at the reds, yellows and blues [...] without the green ones"

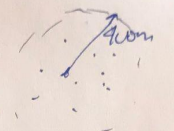
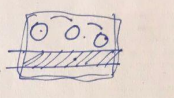


The first consequence is an urban area with three venues that score low on political diversity. This verdict changes when the users are liberated from the venues.

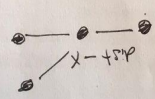


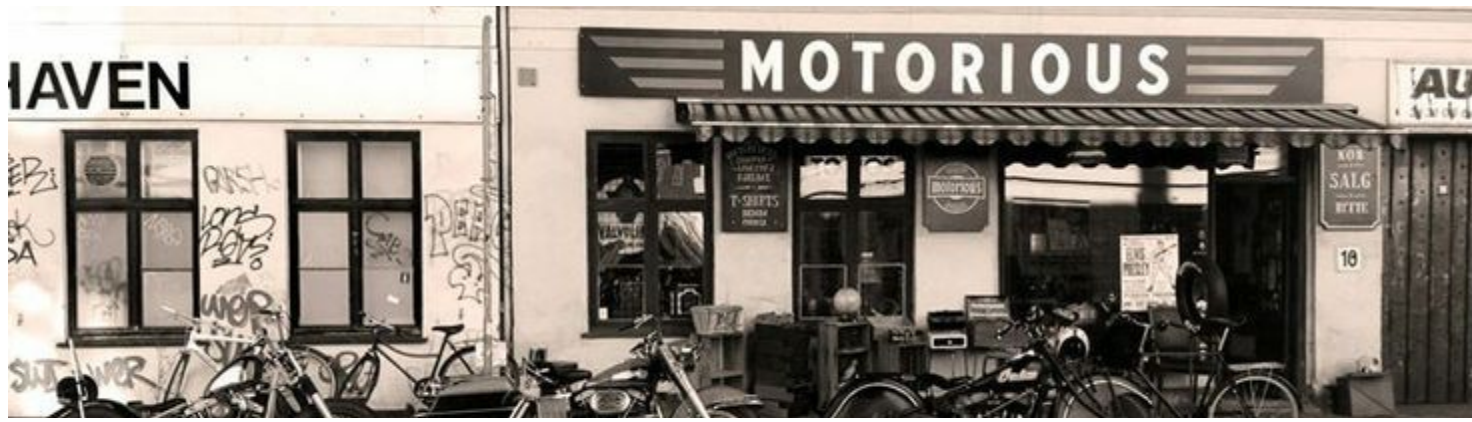
"If that street, [...] had the opportunity to attract [...] red, yellow and blue there, that's super diverse"



• proximity $\frac{0}{0}$ District 
 • density $\frac{0}{m^2}$ street 
 • intensity  venue
 Damhuskroen

public space
 public life
 program
 • music
 • tattoos
 • holiday
 Design program Life





Outside
Gehl's
control

Vs



Within
Gehl's
control

Prompts for your case work and presentations

- How do you ensure that your data project is aligned with the interests of your case companies, given that these interests will likely become much clearer, more precise, explicit and even possible redefined when they see your presentation?
- How will you ensure that the new questions and hypotheses that can be discovered by exploring your data are also vetted and prioritized by your case companies?
- How will you ensure that your case companies do not impose the interpretations that align with their preconceived ideas on your results?